

## 8. Perform email analysis using the Autopsy tool

**Aim:** To analyze and extract email evidence from a forensic image or data source using the **Autopsy** tool — identifying email headers, sender/receiver details, attachments, and message content for digital investigation.

### Theory

**Autopsy** is a digital forensic tool that helps investigators analyze hard drives, mobile devices, and disk images.

It provides modules for:

- File system analysis
- Email and web artifacts recovery
- Timeline analysis
- Keyword search and reporting

**Email forensics** involves the study of email messages to:

- Identify sender/receiver information
- Trace message routes
- Recover deleted or hidden emails
- Extract attachments and metadata

Autopsy supports **email analysis** mainly from:

- **Mozilla Thunderbird (.mbox)** files
- **Microsoft Outlook (.pst / .ost)** files
- **E-mail archives extracted from disk images**

### Requirements

- **Autopsy** tool installed
  - (Windows: from <https://www.autopsy.com>)
  - (Kali Linux: `sudo apt install autopsy`)
- Sample **email archive or forensic image**
  - Example: .pst, .ost, .mbox, or .img file

### Procedure

1. **Open Autopsy**
  - Launch Autopsy from the Applications menu or run:
  - `autopsy`
  - Open in a web browser at: `http://localhost:9999/autopsy`
2. **Create a New Case**
  - Click **“New Case”**
  - Enter Case Name (e.g., *Email\_Analysis\_Case*)
  - Set Base Directory (e.g., `/cases/email_analysis`)
  - Click **Next** → **Finish**
3. **Add Data Source**
  - Choose **“Add Data Source”**
  - Select the type:

- **Disk Image or VM File** (for .img, .dd, etc.)
  - **Logical Files** (for .pst, .mbox, etc.)
- Browse and add your email file.
- 4. **Select Ingest Modules**
  - Enable the following modules:
    - **File Type Identification**
    - **Email Parser**
    - **Keyword Search**
    - **Metadata Extractor**
  - Click **Next** → **Finish** to start analysis.
- 5. **View Email Artifacts**
  - In the **Tree View (left panel)** → expand “**Email Messages**” or “**Results**” → “**Email**”
  - You’ll see:
    - Sender
    - Recipient
    - Subject
    - Date/Time
    - Message Body
- 6. **Analyze Headers**
  - Select an email → view **Header details**
  - Note fields like:
    - *From, To, CC, BCC*
    - *Message-ID*
    - *Return-Path*
    - *Received path* (used to trace source IP)
- 7. **Inspect Attachments**
  - Under “**Email Attachments**”, extract or open any attached files.
  - Verify potential malware, document metadata, or hidden data.
- 8. **Generate Report**
  - Go to “**Generate Report**”
  - Choose **HTML / CSV format**
  - Select **Email Messages** to include.
  - Save report (e.g., Email\_Report.html).

## Observations

| Field        | Example Output       |
|--------------|----------------------|
| Sender       | john.doe@example.com |
| Recipient    | alice@example.org    |
| Subject      | “Invoice Details”    |
| Message Time | 2025-10-07 15:32:10  |
| Source IP    | 192.168.42.120       |
| Attachment   | invoice.pdf          |

## Result

Successfully analyzed an email archive using the **Autopsy** tool. Extracted metadata, sender/receiver info, message content, and attachments for forensic reporting.

## Viva Questions

1. What is the purpose of email forensics?
2. What file formats are supported by Autopsy for email analysis?
3. What information can you get from email headers?
4. What is the difference between .pst and .mbox files?
5. How can attachments be used as digital evidence?

## If Autopsy Not Have followed Steps:-

### Step 1 — Check if Autopsy is already installed

Run this in terminal:

```
autopsy -v
```

#### If Installed:

You'll see something like:

Autopsy 4.x.x

Then you can start it using:

```
autopsy
```

and open your browser → <http://localhost:9999/autopsy>

#### If Not Installed:

You'll see:

Command 'autopsy' not found

Then move to Step 2.

### Step 2 — Update Kali repositories

Before installing, update your system:

```
sudo apt update && sudo apt upgrade -y
```

### Step 3 — Install Autopsy

Run:

```
sudo apt install autopsy -y
```

This installs **Autopsy** and its dependencies (Perl scripts, Sleuth Kit, etc.).

### Step 4 — Verify installation

Once installed, check version:

```
autopsy -v
```

Or locate it:

```
which autopsy
```

You should see output like /usr/bin/autopsy.

### Step 5 — Launch Autopsy

To start:

```
sudo autopsy
```

You'll see:

Starting Autopsy Server at port 9999

Then open a browser and go to:

<http://localhost:9999/autopsy>

Now you can create a new case and start analyzing evidence.

### Step 6 — (Optional) Install Sleuth Kit GUI

Autopsy works with **The Sleuth Kit**, which is also useful for forensic analysis:

```
sudo apt install sleuthkit -y
```

You can verify:

```
tsk_recover -V
```

```
(kali㉿ kali)-[~]
$ sudo autopsy

=====

Autopsy Forensic Browser
http://www.sleuthkit.org/autopsy/
ver 2.24

=====

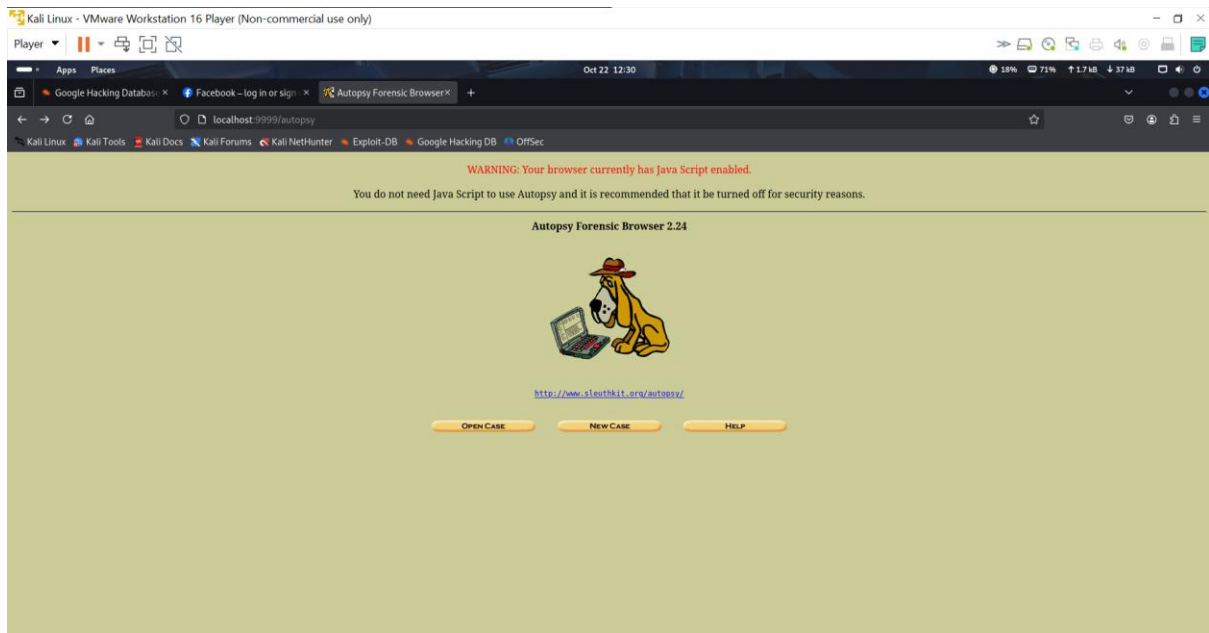
Evidence Locker: /var/lib/autopsy
Start Time: Wed Oct 22 12:17:32 2025
Remote Host: localhost
Local Port: 9999

Open an HTML browser on the remote host and paste this URL in it:

http://localhost:9999/autopsy

Keep this process running and use <ctrl-c> to exit
```

Then Open Vm Ware and open Firefox Browser, open <http://localhost:9999/autopsy>



Then Click **NewCase**

